OpenJLS

JPEG-LS Lossless Encoder IP Core — Datasheet

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1 Overview

OpenJLS is a JPEG-LS lossless encoder for FPGAs. It encodes single-component (grayscale) images at one pixel per clock with no external memory, streaming a standards-compliant JPEG-LS bitstream out over a standard ready/valid streaming handshake. Source code, license and the full verification suite are public at `https://github.com/VitorMendesC/OpenJLS`.

Table 1: Key specifications.

Standard	JPEG-LS (ISO/IEC 14495-1, ITU-T T.87)
Compression	Lossless only
Components	Single (grayscale)
Pixel depth	8–16 bits (`BITNESS`)
Image size	Up to 65535 × 65535 px
Throughput	1 pixel / clock
Latency	14 clock cycles
f_{max}	∼240 MHz (Xilinx `xczu7eg-1`)
Interface	Ready/valid streaming (AXI4-Stream / Avalon-ST compatible)
Memory	On-chip line buffer, one image row; no external RAM
Resources	∼8k LUT, ∼2.1k FF + BRAM scaling with image width
Portability	Vendor-agnostic VHDL

Revision history

Table 2: Revision history.

Version	Date	Notes
1.0	2026-06	Initial release.

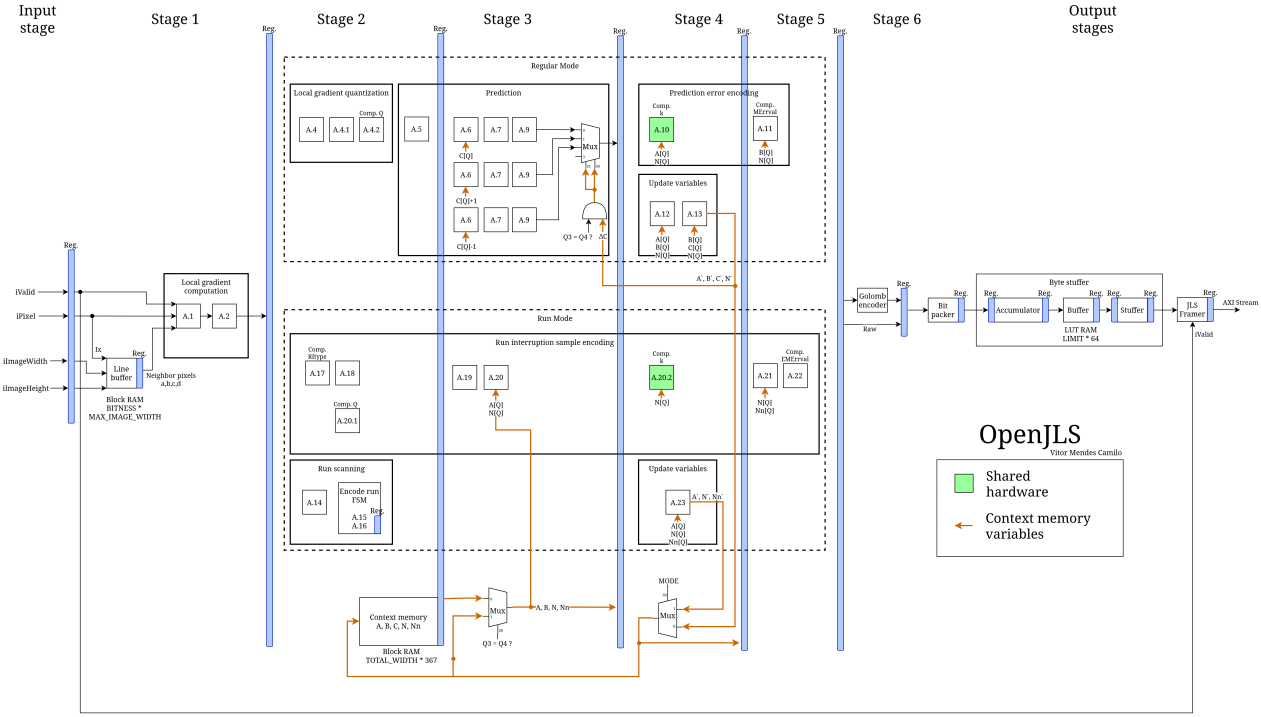
2 Functional description

The core is a fully pipelined encoder with a latency of 14 clock cycles: an input register, six processing stages (gradient computation, context modeling, MED prediction with bias correction, Golomb–Rice / run-length coding) and three internally registered output stages (bit packer, byte stuffer, framer).

End-of-image is derived internally from the dimensions provided at the ports `iImageWidth` and `iImageHeight`. The core never stalls — neither at EOI (end of image) nor at EOL (end of line) — which allows processing multiple images back-to-back. The only exception is when the downstream stalls long enough for the byte-stuffer buffer to fill, which in turn stalls the upstream.

While the pipeline latency is 14 cycles, the core does not emit an output beat every cycle: the output stages accumulate compressed bits and assert `oValid` only once a full `OUT_WIDTH` word is ready. As the bitstream is compressed, these output beats are infrequent relative to the one-pixel-per-clock input.

Figure 1: Encoder pipeline architecture.



A full-resolution version of this diagram is available in the repository: `https://github.com/VitorMendesC/OpenJLS`.

3 Generics

The core is configured by the four generics in Table 3.

Table 3: Configuration generics.

Generic	Range	Description
`BITNESS`	8–16	Pixel bit depth.
`MAX_IMAGE_WIDTH`	4–65535	Largest width supported; sets line-buffer depth.
`MAX_IMAGE_HEIGHT`	1–65535	Largest height supported.
`OUT_WIDTH`	48–1024	Output bus width in bits; multiple of 8.

`MAX_IMAGE_WIDTH` and `MAX_IMAGE_HEIGHT` fix the *compile-time* maximum image size: they size the on-chip line buffer and the dimension counters, so a larger maximum costs more BRAM. They do not select the size of a given image — the dimensions of each encoded image are chosen at *run time* through the `iImageWidth` and `iImageHeight` ports (Table 4), which accept any value from the minimum up to the configured maximum.

4 Ports

All ports are synchronous to the rising edge of `iClk` — the core is a single clock domain. The streaming ports use a plain ready/valid handshake; in Table 4 the **AXIS** column gives the one-to-one AXI4-Stream signal mapping for that ecosystem (Avalon-ST maps the same way at `readyLatency` = 0).

Table 4: Port list and AXI4-Stream mapping.

Port	Dir	Width	AXIS	Role
`iClk`	in	1	—	Clock.
`iRst`	in	1	—	Synchronous reset, active high; latches dimensions.
`iImageWidth`	in	$\lceil \log_2(\text{MAX\_IMAGE\_WIDTH}+1) \rceil$	—	Image width, pixels (configuration).
`iImageHeight`	in	$\lceil \log_2(\text{MAX\_IMAGE\_HEIGHT}+1) \rceil$	—	Image height, pixels (configuration).
`iValid`	in	1	`TVALID`	Pixel valid.
`iPixel`	in	`BITNESS`	`TDATA`	Pixel data.
`oReady`	out	1	`TREADY`	Input ready.
`oData`	out	`OUT_WIDTH`	`TDATA`	Bitstream beat.
`oValid`	out	1	`TVALID`	Output valid.
`oKeep`	out	`OUT_WIDTH/8`	`TKEEP`	Valid-byte mask on final beat.
`oLast`	out	1	`TLAST`	End of image.
`iReady`	in	1	`TREADY`	Downstream ready.

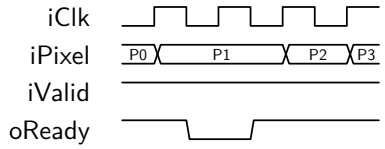
IMPORTANT:

- `iImageWidth` and `iImageHeight` are only sampled while `iRst` is asserted (`'1'`).
- If an invalid value is provided to `iImageWidth`, that is `iImageWidth > MAX_IMAGE_WIDTH` or `iImageWidth < 4`, the encoder uses `MAX_IMAGE_WIDTH` as the image width.
- If an invalid value is provided to `iImageHeight`, that is `iImageHeight > MAX_IMAGE_HEIGHT` or `iImageHeight < 1`, the encoder uses `MAX_IMAGE_HEIGHT` as the image height.
- If you don't intend to change the image dimensions at run-time, wire both inputs to `0` so that the maximum values are used.
- The ports are only $\lceil \log_2(\text{MAX}+1) \rceil$ bits wide, so a value far above the maximum can overflow the port and wrap back into the valid range; the encoder cannot detect this case and won't substitute the maximum.

5 Timing and protocol

Pixel input. Pixels are streamed in scan order — the first row left to right, then the second row, and so on — each an unsigned value on `iPixel`. A pixel transfers on every cycle where `iValid` and `oReady` are both high. The core asserts `oReady` continuously and deasserts it only when the FIFO in byte stuffer fills under downstream backpressure. Figure 2 shows a transfer with a brief downstream stall.

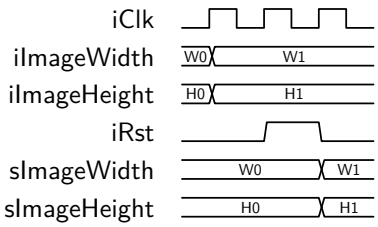
Figure 2: Pixel-input handshake.



In Figure 2, `P2` is held flat across the stall and accepted when `oReady` returns high; no pixel is dropped while `oReady` is low.

Reset and configuration. `iRst` is level-sensitive (hold it high for at least one clock) and clears the pipeline. The image dimensions are captured only while `iRst` is high: present `iImageWidth` and `iImageHeight`, hold them stable across the reset pulse, and they latch on the rising edge (Figure 3). A reset is therefore needed only to capture the dimensions — before the first image and whenever the image size changes; with the size unchanged, images encode back-to-back.

Figure 3: Reset and dimension capture.



In Figure 3, the new size (`W1`/`H1`) is driven for several cycles, then `iRst` pulses high; the dimensions register on the rising edge while `iRst` is high.

Output. Beats appear on `oData`/`oValid` subject to `iReady`; `oLast` marks the final beat (EOI) and `oKeep` flags its valid bytes. The bitstream is byte-serial and MSB-first: the earliest byte occupies the most-significant byte of `oData`. Pipeline latency is 14 clock cycles.

6 Resources and performance

Characterized on Xilinx Zynq UltraScale+ `xczu7eg-fbvb900-1-e`, Vivado 2025.2, using the generics `BITNESS = 12` and `OUT_WIDTH = 64`, RTL-only with no floorplanning. f_{max} is the true maximum, found by over-constraining the clock until timing fails. At one pixel per clock, ∼240 MHz is ∼240 Mpixel/s.
Figure 4 and Table 5 report f_{max} across `MAX_IMAGE_WIDTH`; Figure 5 and Table 6 report the resource usage.

Figure 4: Maximum frequency vs maximum image width.

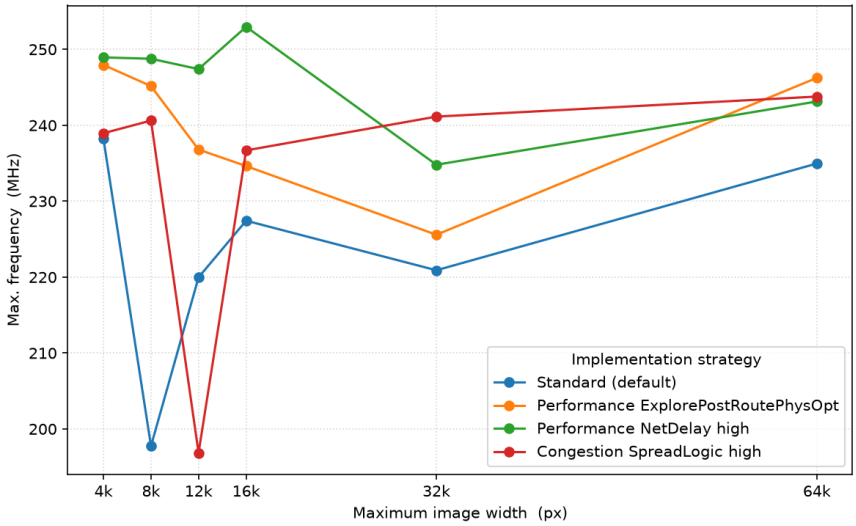


Table 5: Maximum frequency by maximum image width and strategy.

`MAX_IMAGE_WIDTH`	Default	EPRPO	NetDelay	Congestion
4096	238.2	247.9	**248.9**	238.9
8192	197.8	245.2	**248.8**	240.6
12288	220.0	236.8	**247.4**	196.8
16384	227.4	234.6	**253.0**	236.7
32768	220.9	225.6	234.8	**241.1**
65535	235.0	**246.2**	243.1	243.8

Figure 5: Resource usage vs maximum image width.

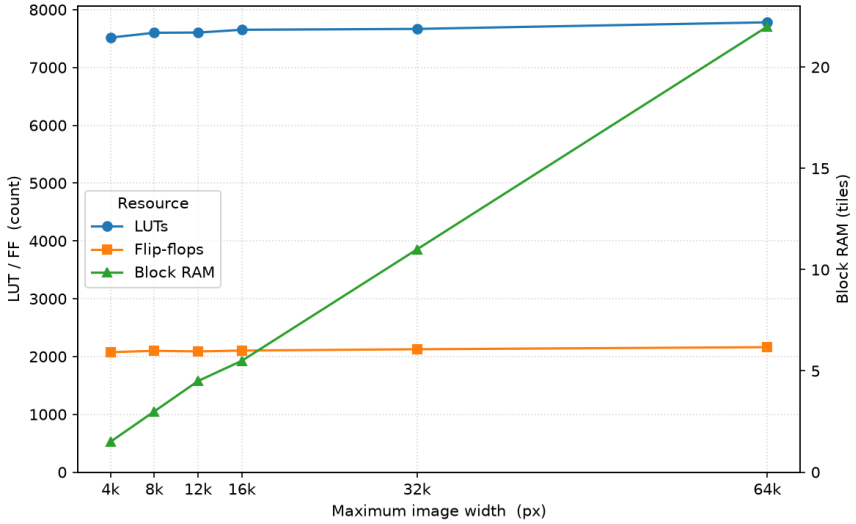


Table 6: Resource usage by maximum image width.

MAX_IMAGE_WIDTH	LUT	FF	BRAM
4096	7517	2075	1.5
8192	7600	2099	3.0
12288	7604	2088	4.5
16384	7652	2103	5.5
32768	7667	2126	11.0
65535	7781	2162	22.0

In Table 5 frequencies are in MHz and the best strategy per row is in bold; the strategies are `Performance_ExplorePostRoutePhysOpt` (EPRPO), `Performance_NetDelay_high` (NetDelay) and `Congestion_SpreadLogic_high` (Congestion). The design is congestion-bound, so the winning strategy shifts with image size; a given netlist is deterministic, but the per-size winner is placement-sensitive and can shift when the netlist changes, so the best-of band (∼241–253 MHz) is the headline figure. Table 6 is for the default strategy; utilization is near-identical across strategies. Logic is essentially constant with image size; only the line-buffer BRAM scales with width and pixel depth.

7 Integration example

```
u_enc : entity work.openjls_top
  generic map (
    BITNESS => 8,
    MAX_IMAGE_WIDTH => 4096,
    MAX_IMAGE_HEIGHT => 4096,
    OUT_WIDTH => 64 )
  port map (
    iClk => clk,
    iRst => rst,
    iImageWidth => img_w, -- latched on reset
    iImageHeight => img_h, -- latched on reset
    iValid => s_valid, -- pixel stream (AXIS slave)
    iPixel => s_data,
    oReady => s_ready,
    oData => m_tdata, -- bitstream (AXIS master)
    oValid => m_tvalid,
    oKeep => m_tkeep,
    oLast => m_tlast,
    iReady => m_tready );
```

Being a single entity with standard ready/valid ports, `openjls_top` can equally be dropped onto a block diagram and wired there, instead of instantiating it in HDL.

8 Conformance and verification

Verified by simulation with NVC using a layered, self-checking suite run at both RTL and post-synthesis (gate-level) stages:

- **OSVVM** — per-module tests against independent T.87-derived reference models, plus top-level control-plane stress (reset injection, backpressure, randomized sizes) with requirements tracking.
- **Coverage** — OSVVM functional coverage with NVC structural code coverage (99%+ statements).
- **Golden model** — byte-exact vs the independent CharLS C++ encoder over 287 real and synthetic images, plus ISO/IEC 14495-1 vectors.
- **Design contracts** — embedded PSL assertions (ready/valid handshake protocol, internal handshakes) on every run.
- **Post-synthesis** — the stress test and a golden-model subset re-run on the synthesized netlist.

Table 7 is the latest regression snapshot (live, per-test reports at `https://vitormendesc.github.io/OpenJLS/`).

Table 7: Latest regression snapshot.

Suite	Status	Test/Cov	Summary
OSVVM suite	PASS	100%	36 tests, 129 810 affirmations (module + top control-plane)
NVC code coverage	info	99.3%	Per-file statement breakdown
Golden model	PASS	100%	287/287 images byte-exact vs CharLS
Post-synth OSVVM	PASS	100%	Control-plane stress on the gate-level netlist
Post-synth golden model	PASS	100%	156/156 images byte-exact vs CharLS

References. ITU-T T.87 / ISO/IEC 14495-1; Y. M. Mert, *Key Architectural Optimizations for Hardware Efficient JPEG-LS Encoder*, IEEE 2018; CharLS; open-logic.